Comprehensive: Fractions + Decimals + Equations

Semester 1Comprehensive Review integrated lesson • 75 minutes • 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Volume A Unit 3 (Fractions) + Unit 4 (Decimals) + Unit 5 (Simple Equations)

Integration scope: Class 7-10 (Fraction Operations) + Class 12-13 (Decimal Operations) + Class 14-15 (Simple Equations)

Core trap:□ T2 order of operations + T9 fraction operation + T3 equation trap 🔴 SSPA cross-topic hybrid

SSPA correlation: 🔴 High frequency 20-25%, cross-topic mixed questions are the key to winning stars in Paper 2

Prerequisite knowledge: Common fractions, addition and subtraction of fractions with different denominators • Four arithmetic operations with decimals • One-step/two-step equation solution

Course objectives:❶ Unified format strategy for mixed calculations of fractions and decimals ❷ Handling of fractions/decimals in equations ❸ Comprehensive application across topics

Student Name: ____ Class: ____ Date: ____ Time Spent: ____

I.Warm-Up Questions（total 5 question，5 minutes）

#	Question	Difficulty	Working Space（Show full working）
1	Convert 0.75 to its simplest fraction.	Basic	
2	Convert $\frac{3}{4}$ to a decimal.	Basic	
3	Solve the equation: $x+2=5$，$x=$?	Basic	
4	Calculation: $\frac{1}{2}+0.5=?$ (Tip: Unify the format)	Advanced	
5	Calculate LCM(4, 5, 10) = ?	Advanced	

II.Core Knowledge + Worked Examples

Knowledge Point 1: Mixed Calculation of Fractions + Decimals - Unified Format Strategy 🔴 SSPA Required Exam

Core principle: Numbers in different forms cannot be mixed directly for operations!

① **Strategy A: Convert all to fractions**— Suitable for situations where the decimal is "a finite decimal and has a simple fraction correspondence"

② **Strategy B: Convert all to a decimal**-- Suitable for situations where the denominator of a fraction is 2, 4, 5, 8, 10, etc. (can be directly divided)

③ **Judgment method**: If the fraction cannot be converted into a finite decimal (such as $\frac{1}{3}$), strategy A (all converted to fractions) must be used

④ **After unified format**: Calculated according to normal fraction or decimal operation rules

Commonly used fractions ↔ Decimals comparison table

1/2 = 0.5	1/4 = 0.25	3/4 = 0.75	1/5 = 0.2	2/5 = 0.4	1/8 = 0.125
1/10 = 0.1	3/5 = 0.6	4/5 = 0.8	3/8 = 0.375	5/8 = 0.625	1/3 ≈ 0.333

⚠ 1/3, 1/6, 1/7, 1/9, etc. cannot be converted into finite decimals → Strategy A (full conversion of fractions) must be used!

Trap detonation example 1 (the most important demonstration in this class - T9 fraction and decimal mixing trap)

calculate: $\frac{1}{2}+0.25=$ **?**

❌ **Common mistakes (60% students)**

$$\frac{1}{2}+0.25=1.25/2\ ???$$

If the formats of fractions and decimals are not consistent, just add them - totally wrong!

✅ **Correct solution (Strategy A: Turn all points)**

$$0.25=\frac{1}{4}\rightarrow\frac{1}{2}+\frac{1}{4}=\frac{3}{4}=0.75$$

Unified into fractions: 0.25=14 → LCM(2,4)=4 → 24+14=34

Example 2 - Demonstration of strategy B (convert all to decimals)

calculate: $\frac{3}{5} + 0.2 = ?$

❌ **Common mistakes**

$$\frac{3}{5} + 0.2 = \frac{3}{7}$$

Randomly adding 0.2 as the denominator - it makes no sense!

✅ **Correct solution (Strategy B: Convert all to decimals)**

$$\frac{3}{5} = 0.6 \rightarrow 0.6 + 0.2 = 0.8 = \frac{4}{5}$$

35=3÷5=0.6, or use strategy A: 35+15=45

Example 3 - Situation when strategy A must be used (fractions cannot be converted into finite decimals)

calculate: $\frac{1}{3} + 0.5 = ?$

🧠 **Tip: "For the same problem of fractions and decimals, do it in a unified format first; the fractions cannot be converted to a limited number, so the most stable formation is to convert all fractions to fractions."**

⚠️ **T9 The most frequent trap: When fractions and decimals are mixed, the format is not uniform - they are directly added as similar numbers!**

⚠️ **T2 Order of operations trap: When there are addition, subtraction, multiplication and division in mixed questions, you must first multiply and divide and then add and subtract!**

Knowledge Point 1 Synchronization practice (write out process in a unified format)

#	Question	Difficulty	Working Space
Example 1	$\frac{1}{4} + 0.5 = ?$	🌱	
Example 2	$0.6 + \frac{2}{5} = ?$	🌱	
Example 3	$\frac{2}{3} + 0.25 = ?$	🌿	

Knowledge point 2: Fractions and decimals in equations 🔴 SSPA required test

Dealing with fractions/decimals in equations - two strategies:

① **Denominator removal method |||SEP|||: Multiply both sides of the equation by LCM, eliminate all denominators → become an integer equation**Removal of decimals |||SEP|||: Multiply both sides of the equation by 10, 100, 1000 → become an integer equation

② **Direct calculation method |||SEP|||: Calculate the fraction/decimal part first, then solve the equation (suitable for simple cases)**Key reminder |||SEP|||: Both sides of the equation must perform the same |||SEP|||
③ **direct calculation method**: Calculate the fraction/decimal part first, then solve the equation (suitable for simple situations)
④ **Key reminder**: Both sides of the equation must be done**same**Operations (equivalence axiom)
⑤ **Root verification**: Substitute into the original equation to check to ensure there are no calculation errors.

Trap detonation example 4 (T3 equation trap - forget to multiply each term after removing the denominator)

Solve the equation: $\frac{x}{2}+\frac{1}{3}=\frac{5}{6}$ and find x。

❌ **Common mistakes (70% students)**

only ride $\frac{x}{2} \rightarrow 3x+\frac{1}{3}=\frac{5}{6}$

Forget that each term on both sides must be multiplied by LCM(2,3,6)=6!

✅ **Correct solution**

$\times 6 \rightarrow 3x+2=5 \rightarrow 3x=3 \rightarrow x=1$

Multiply both sides by 6: 6×x2+6×13=6×56 → 3x+2=5 → x=1

Example 5 - Equations containing decimals (decimal removal method)

Solve the equation: $0.5y+0.3=1.3$, find y。

Example 6 - Mixed numbers in equations (fractions + decimals co-occur)

Solve the equation: $x+\frac{1}{2}=1.5$, find x。

🧠 **Tip: "If the equation has a denominator, multiply both sides by LCM; if the equation has a decimal, multiply both sides by 10^n; multiply everything, so you don't miss anything!"**

⚠ **T3 The most frequent trap: When removing denominators/decimals, only multiply terms containing fractions/decimals and forget to multiply constant terms!**

⚠ **T2 Trap: When the equation contains parentheses, remove the parentheses first and then the denominator. The order cannot be messed up!**

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
Example 4	Solve the equation: $\frac{x}{3}+\frac{1}{2}=\frac{5}{6}$ and find x。	🌿	
Example 5	Solve the equation: $0.2y+0.8=1.4$, find y。	🌿	

Knowledge point 3: Comprehensive application across topics 🔴 The key to winning SSPA stars

Comprehensive question-solving framework (four-step cracking method):

① **Classification of reading questions |||SEP|||: Determine what knowledge points are involved in the question (fraction? Decimal? Equation? Area?)**Unified format |||SEP|||: Fractions and decimals cannot coexist → Choose a format to unify the whole question

② **Step-by-step formula**：List the formula first, then solve the equation (if you need to set up unknowns)

③ **Verify the answer**：Return to the original question to check whether it is reasonable + unit/answer sentence is complete

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①	②	③	④
Reading question classification Determine what knowledge points are involved	**unified format** Choose one of fractions/decimals	**step by step formula** Calculation → Equation → Solve	**Verify answer** Check the complete answer

Example 7 - Fraction + Decimal word problem

Xiao Ming has a $\frac{3}{4}$ meter ribbon, and Xiao Mei has a 0.5 meter ribbon. How many meters of ribbon do they have in common? (Answer in fractions)

Example 8 - Fraction + Equation word problems

A bottle of juice, after drinking |||SEP|||, I drank another 0.4 liters, for a total of $\frac{1}{3}$ liters. Assume that the original number is raised to |||SEP|||, make an equation and solve it. $\frac{7}{10}$ Lift. Suppose the original amount is upgraded to x, make equations and solve them.

❌ Common Mistake – Making Wrong Equations

$$x - \frac{1}{3} - 0.4 = \frac{7}{10}$$

Think of "drinking" as subtraction - the question is to find "how much was drunk in total", not to find the remainder!

✅ Correct column formula

$$\frac{1}{3}x + 0.4 = \frac{7}{10} \rightarrow \frac{1}{3}x = \frac{3}{10} \rightarrow x = \frac{9}{10}$$

Always use decimals: 0.4=25=410 → 13x=310 → x=910

🧠 Tip: "For application questions, follow four steps: Read the question → Unify → Column → Verify. Fractions and decimals have different shapes. Count first after unifying."

⚠️ T2+T9+T3 mixed trap: The most common mistake in comprehensive questions is-forgetting the order of operations (multiplication and division first, then addition and subtraction), and at the same time, calculating fractions and decimals without unification!

III. Lesson Layered Synchronization Practice

Basic layer (total 6 questions, all must do - unified format basic function)

#	Question	Difficulty	Working Space
6	$0.5+\dfrac{1}{4}=?$	🌱	
7	$\dfrac{3}{10}+0.7=?$	🌱	
8	$0.2+\dfrac{3}{5}=?$	🌱	
9	Solve the equation:$x+0.3=0.9$，$x=?$	🌱	
10	Solve the equation:$\frac{x}{4}=0.5$，$x=?$	🌿	
11	$\frac{1}{2}+\frac{2}{5}+0.1=?$	🌿	

Advanced layer (total 6 questions, 🚶🚀 choose do — equation+mixed operations)

#	Question	Difficulty	Working Space
12	Solve the equation:$\frac{x}{3}+\frac{1}{6}=\frac{1}{2}$，$x=?$	🌿	
13	Solve the equation: $0.4y-0.1=0.7$，$y=?$	🌿	
14	$\frac{3}{4}-0.25+\frac{1}{8}=?$	🌿	
15	Solve the equation:$\frac{2x}{5}=0.8$，$x=?$	🌳	
16	$0.6\times\frac{1}{3}+\frac{1}{2}=?$ (note the order of operations)	🌳	
17	Solve the equation:$x+\frac{1}{4}=0.75$，$x=?$	🌿	

🌳 challenge layer (total 6 questions, 🚀 choose do — Cross-Topiccomprehensiveapplication)

#	Question	Difficulty	Working Space
18	Solve the equation:$\frac{x}{2}+\frac{x}{3}=\frac{5}{6}$，$x=?$	🌳	
19	$\frac{3}{8}+0.375+\frac{1}{4}-0.5=?$	🌳	
20	Solve the equation:$\frac{3}{4}x-0.25=\frac{1}{2}$，$x=?$	🌳	

21	Area of triangle = $\frac{1}{2}$ × base × height. It is known that the area = 12 cm^2, the height = 4.8 cm, let the base be b cm, and calculate the equation b。		
22	Solve the equation: $0.5(x+2) = \frac{3}{4} + \frac{1}{4}$，$x =$?		
23	A rope is $5\frac{1}{2}$ meters long. After cutting off 1.75 meters, the remaining length is divided into 3 equal parts. Assume that the length of each segment is y meters, and calculate the equation y。		

IV.applicationquestionspecial topic training (total 8 questions)

Life application question (all must do, column → unified format → calculate → answer sentence)

#	Question	Difficulty	Working Space
24	A bottle of water has 1.5 liters. I drank $\frac{1}{4}$ liters in the morning and 0.5 liters in the afternoon. How many liters did you drink throughout the day? (Answer in fractions)		
25	The cake shop sold $\frac{3}{5}$ cakes in the morning and 0.4 cakes in the afternoon. How many cakes were sold throughout the day? (Answer in decimals)		
26	Xiao Ming weighs 35.5 kg, and Xiao Qiang weighs $\frac{3}{10}$ kg less than Xiao Ming. How many kilograms does Xiaoqiang weigh? (Answer in fractions)		
27	The original length of the rope is x meters. After using $\frac{1}{2}$ meters, the remaining length is 2.5 meters. How many meters long was it originally? (Solving equations)		
28	The warehouse stores $\frac{3}{4}$ tons of rice, ships in 0.5 tons, and then ships out $\frac{2}{5}$ tons. How many tons of rice are there in the final warehouse?		
29	The rectangle is 3.5 meters long and $\frac{4}{5}$ meters wide. How many square meters is the area? (Answer in decimals)		
30	Original books in the library n After purchasing $0.2n$ copies, there are now 1,200 books. Make an equation to find the original number of books.		
31	Area of triangle = $\frac{1}{2}$ × base × height. Known base = 8.4 cm, height = $\frac{5}{2}$ cm. Find the area. (Answer in fractions)		

V. ⛰ Ultimate challenge area (total 3 question)

#	Question	Difficulty	Working Space
⛰ 1	Solve the equation: $\frac{x}{2}+\frac{x}{3}+\frac{x}{6}=\frac{11}{2}$ and find |||SEP|||. (an equation with three fractional terms)x. (an equation with three fractional terms)	⛰	
⛰ 2	A has $\frac{3}{5}$ kilograms of candies, B's candies are 0.5 times that of A, and C's candies are $\frac{1}{4}$ kilograms more than A's. How many kilograms of candy do the three of them have in total? (Fractional answer, three steps)	⛰	
⛰ 3	The original water in the water tank L liters. After releasing the |||SEP|||, add another 2.5 liters, now you have 8 liters of water. Assume that the original water is $\frac{1}{3}$ liters and solve the equations. (Hint: Let go L which is what' s left $\frac{1}{3}$ That is what is left $\frac{2}{3}$)	⛰	

VI.Class afterhomework

Basic must-do questions (total 5 questions, mustwrite out unified format step)

#	Question	Difficulty	Working Space
H1	$\frac{1}{2}+0.25=?$	🌱	
H2	$0.6-\frac{1}{5}=?$	🌱	
H3	Solve the equation: $\frac{x}{3}=\frac{1}{2}$, $x=?$	🌱	
H4	Solve the equation: $0.3y=0.9$, $y=?$	🌱	
H5	$\frac{3}{4}+0.5+\frac{1}{8}=?$	🌿	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	Solve the equation: $\frac{x}{4}+0.25=\frac{3}{4}$, $x=?$	🌳	
H7	A number n where $\frac{2}{5}$ is equal to 0.6. beg n。	🌳	
H8	Solve the equation: $\frac{2}{3}x-0.5=\frac{1}{6}$, $x=?$	⛰	

VII. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point ☐ I can complete 🌱basic questions independently

☐ I can challenge 🌿advanced questions ☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall ☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge 🌿Advanced questions (80%+ correct) ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**Mixed fractions and decimals are not in a unified format**： $\frac{1}{2}$+0.3 Just add them randomly	First unify the format: convert all fractions or convert all decimals
2	**When removing the denominator, the constant term is missing |||SEP|||: ×LCM only multiplies the fractional term**:×LCM only multiplies fractional terms	Each term of the equation must be multiplied by LCM, including purely numeric terms
3	**When subtracting decimals, multiply by the wrong multiple |||SEP|||: 0.3 → multiply by 10 instead of 10**:0.3→multiply by 10 instead of 10	See the most decimal places: 0.5→×10, 0.25→×100
4	**Mixed operations forget to multiply and divide first and then add and subtract** (T2)	When there is × ÷, do multiplication and division first, then addition and subtraction
5	**Fractions cannot be converted into finite decimals but are hard converted**： $\frac{1}{3}$=0.333...	When encountering infinite decimals→you must use the full fraction strategy
6	**Forgetting to change the sign when transferring terms in an equation** (T3)	Shift term = add/subtract both sides at the same time, the sign should change accordingly
7	**Missing sentences/units in application questions**	Must write "Answer:..." + complete unit, step points will be deducted